

LAB④ Mini Project - SRS DocumentProject Title: GenAI Education Assistant4 IntroductionPROBLEM STATEMENT:

Students lack personalized, adaptive learning support; existing tools fail to provide interactive, context-aware guidance & real-time doubt resolution.

1. Introduction:1.1 Purpose

Provide students with an intelligent education assistant powered by GenAI. Develop adaptive learning through Q&A, concept explanations & personalized tests & quizzes.

1.2

Scope:

It will track knowledge using student performance history. It will generate personalized recommendation for next topic or quiz.

1.3

Overview:

A web application connected to AI backend. Includes login, Q&A chat, quizzes & dashboard.

2. General Description :

Standalone webapp integrated with AI API & database. It is dependent on internet, API, usage cost. The application is built on RAG (Retrieval-Augmented-Generation) architecture. It is designed to avoid any kind of hallucinations that might come with the use of AI assistant.

3. Functional Requirements

3.1 User registration & login

3.2 Personalized quiz generation

3.3 KT for progress tracking

3.4 Recommendation engine for next topic.

4. Interface Requirement

4.1 UI : ChatBot Interface

4.2 Dashboard with charts & progress bars

4.3 Hardware: Runs on PC/Mobile with internet

5. Performance Requirements

5.1 Q&A Response time < 3 sec

5.2 Support 500 concurrent users

5.3 @ a time

5.3 Uptime: 95%

6. Design Requirements:

- 6.1 Frontend : React / Flutter
- 6.2 Backend : Django , FastAPI
- 6.3 Database : SQL / MongoDB
- 6.4 AI Integration.
- 6.4 DevOps : Docker & Kubernetes

7. Non-Functional Attributes

- 7.1 Usability
- 7.2 Reliability
- 7.3 Scalability
- 7.4 Security

8. Preliminary Budget:

Phase	Duration	Cost Estimate
Req. Analysis	1 week	Rs. 500
System Design	2 week	Rs. 1000
Implementation	4 week	Rs. 1000
Testing	2 week	Rs. 100
Deployment	1 week	Rs. 400
Total	10 week	Rs. 3000

UML DIAGRAM

